Software Engineering

Assignment-4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following are the focus of the requirements analysis activity?

a. Weed out ambiguities in the requirements
b. Weed out inconsistencies in the requirements
c. Weed out non-functional requirements
d. Weed out functional requirements
e. Weed out incompleteness in the requirements

Correct Answer: a. Weed out ambiguities in the requirements
b. Weed out inconsistencies in the requirements
e. Weed out incompleteness in the requirements

Detailed Solution:

The main aim of the requirement analysis is to identify all the ambiguities, inconsistencies, and incompleteness and then remove all those ambiguities, inconsistencies, and incompleteness from the requirements.

QUESTION 2:

A software requirements specification (SRS) document should avoid discussing which of the following?

a. Functional requirements
b. Non-functional requirements
c. Design specification
d. Constraints on the implementation
e. Team members and their roles

Correct Answer: c. Design specification
e. Team members and their roles

Detailed Solution:

SRS should not include Designs
–Requirements and designs have different audiences
–Analysis and design are different areas of expertise
Should not include project development plans, team members and their roles. Please refer slide no. 27 of week 4 lecture material.

QUESTION 3:

Consider the following requirement for a word processor software: " The software should provide a facility to convert the file into HTML format.". Which one of the following types of requirements is this?

a. Functional requirement
b. Non-functional requirement
c. Constraint on the implementation
d. Goal of implementation
e. User interface requirement

Correct Answer: a. Functional requirement

Detailed Solution:

Functional requirements describe:
–A set of high-level requirements, each high-level requirement: takes in some data from the user, outputs some data to the user. The question example is a functional requirement where an input taken and convert to desired output. Please refer slide no. 47 – 53 of week 4 lecture material.

QUESTION 4:

Which of the following assertions are **FALSE** about functional requirements?

a. A functional requirement is also called a behavioural requirement
b. A functional requirement should specify the overall system reliability for safety-critical systems
c. A functional requirement may consist of several scenarios of execution
d. A functional requirement is a statement of how certain inputs to program are mapped to the outputs
e. One of the functional requirements may express the programming language that should be used to develop the software

Correct Answer: b. A functional requirement should specify the overall system reliability for safety-critical systems

e. One of the functional requirements may express the programming language that should be used to develop the software

Detailed Solution:

System reliability and specification of programming languages is part of non-functional requirement. So, option **b**. and **e.** are the correct choices.

QUESTION 5:

Consider the following partial description of the security software of an educational institute. "A camera at the main gate should detect an incoming vehicle, and the gate should be opened only for the registered vehicles." Which one of the following types of requirements is this?

a. Functional requirement
b. Non-Functional requirement
c. External interface
d. Design constraint
e. User interface issue

Correct Answer: a. Functional requirement

Detailed Solution:
From the question statement, we can understand that it is an example of functional requirement. Here the function is:
F01: Detect an incoming vehicle
F02: Open gate for registered vehicles

Please refer study material for week-4 (Slide no: 42 and 46-52) for more example.

QUESTION 6:

Which one of the following best describes a functional requirement of a software system?

a) The system must be developed using Python and Django.
b) The system should respond to user input within 2 seconds.
c) The system shall allow users to log in using their email and password.
d) The system must be maintainable and scalable for future enhancements.
e) The system interface must be intuitive and user-friendly.

Correct Answer: c) The system shall allow users to log in using their email and password.

Detailed Solution:
A functional requirement specifies what the system should do — the specific behavior or functions of the system.
Option c describes a specific functionality (user login) that the system must provide — a clear functional requirement. GUI, software interface, user interface etc. are fall under external interface. A web browser provides a user interface to browse internet. So, option d is correct.

Please refer slide no. 35 and 40 of week 4 lecture material for more understanding

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QUESTION 7:

Suppose a software project is underway using the waterfall model of development. When is the requirement specification activity carried out?

a. During the requirements gathering activity
b. Immediately after the requirements analysis activity is complete
c. Before the requirements analysis activity
d. Before the requirements gathering activity
e. Immediately after the feasibility study activity is complete

Correct Answer: b. Immediately after the requirements analysis activity is complete

Detailed Solution:
Please refer the below to understand the answer:

After the analysis requirement specification activity carried out.

QUESTION 8:

Which one of the following is not a requirement gathering technique?

a. Task analysis
b. Scenario analysis
c. Form analysis
d. Review of similar software
e. SRS document review
f. Requirements write-up given by the customer

Correct Answer: e. SRS document review

Detailed Solution:

SRS document review is an activity of SRS document making. After the specification process review take place. So, SRS document review is not a requirement gathering technique. Please refer slide no. 1 to 8 of the week 4 lecture material for more understanding.

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QUESTION 9:

Elimination of which of the following problems in the gathered requirements is not the focus of the requirements analysis task?

a. Unstructuredness
b. Inconsistency
c. Ambiguity
d. Incompleteness
e. Incompactness

Correct Answer: a. Unstructuredness
e. Incompactness

Detailed Solution:
Requirement analysis detect and removes any ambiguity, inconsistency, and incompleteness. So, option **a**. and **e**. are the correct options. Please refer slide no. 9 to 19 of the week 4 lecture material.

QUESTION 10:

Which one of the following views of the software to be developed is implicit in an SRS document?

a. Black box
b. White box
c. Grey box
d. Glass box

Correct Answer: a. Black box

Detailed Solution:

SRS document is known as black-**box specification**:
–The system is considered as a black box whose internal details are not known.
Please refer slide no. 22 of the week 4 lecture material.

**********END**********